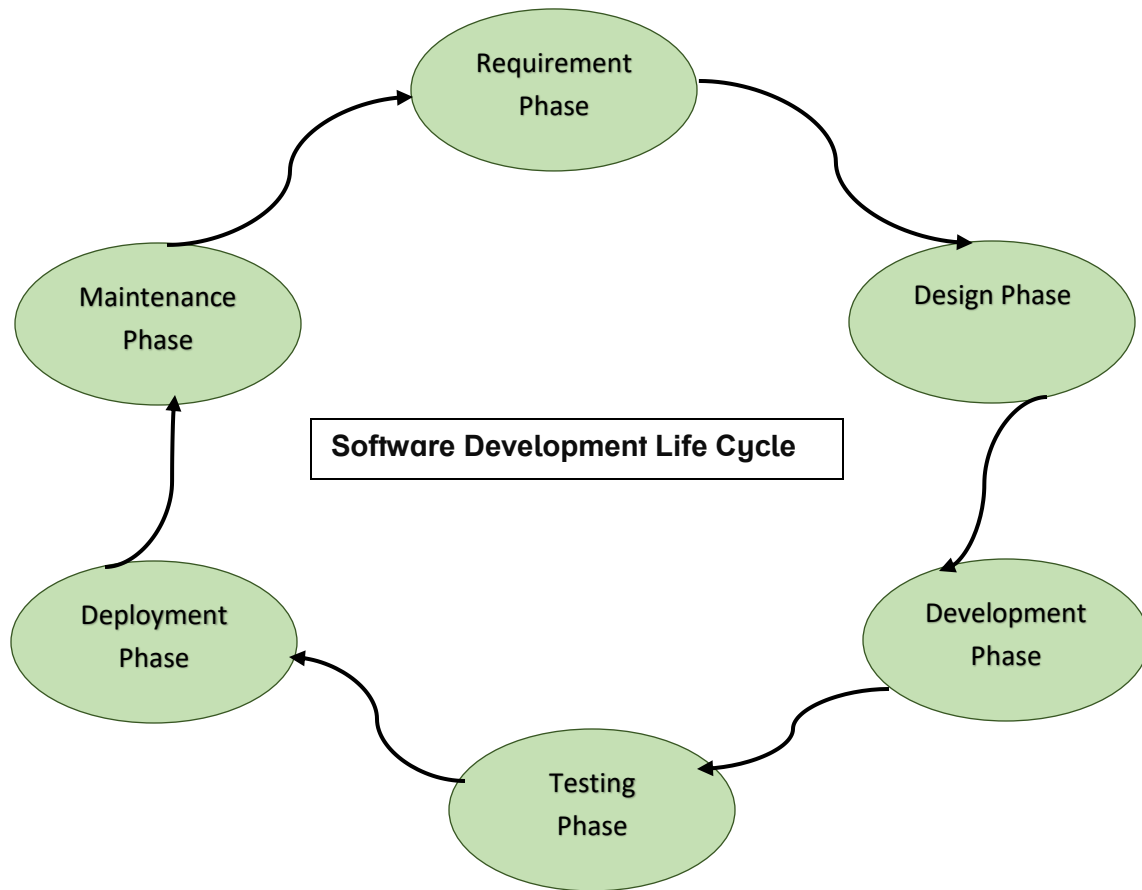


Q1: Draw the SDLC lifecycle diagram and briefly explain the workflow.



Requirement Analysis Phase: Requirement Analysis is the first and one of the most important phases of SDLC. During this phase, the development team communicates with clients and stakeholders to understand their needs and expectations. The team gathers business and user requirements, defines functional and non-functional requirements, identifies system constraints, and prepares the necessary documentation. The main documents include BRD, SRS, Use Cases, and RTM. Finally, the requirements are reviewed, validated, and approved by the stakeholders before moving to the next phase.

Design Phase: Design Phase is the phase where the requirements gathered during the Requirement Analysis phase are converted into a technical blueprint for the software. During this phase, the team decides how the system will work, look, and be structured. It includes designing the system architecture, database, UI/UX, modules, interfaces, and technology stack. The team also defines how different components will interact with each other. The main goal is to create a clear design that developers can follow during the Development phase.

Development Phase: Development Phase is the phase where the actual software is built based on the requirements and design. During this phase, developers write, integrate, and review the code to implement different features and functionalities. The developed modules are combined to create the complete system, and basic checks are performed to ensure the code works as expected. The main goal of this phase is to convert the approved design into a functional software product that is ready for testing.

Testing Phase: Testing Phase is the phase where the developed software is systematically checked to ensure it works as expected and meets the specified requirements. During this phase, QA engineers execute test cases, identify and report defects, verify fixes, and perform regression testing. Different types of testing, such as functional, integration, system, and performance testing, may be performed. The main goal of this phase is to ensure the software is stable, reliable, and ready for deployment.

Deployment Phase: Deployment Phase is the phase where the tested and approved software is released to the production environment for real users. During this phase, the team installs and configures the application, performs final checks, and makes it available to users. Deployment can be done all at once or gradually depending on the project. The main goal is to successfully deliver the software to the end users.

Maintenance Phase: Maintenance Phase is the final phase of the SDLC, where the software is monitored and supported after deployment. During this phase, the team fixes bugs, improves performance, adds new features, and makes necessary updates based on user feedback or changing requirements. The main goal is to keep the software stable, secure, reliable, and useful over time.

Q2: Practical AI exercise-> Generate 5 user stories for an E-commerce application, convert 1 user story into acceptance criteria, AI – Assisted Sprint Backlog, Review AI generated output and improve manually.

Selected AI Tool: ChatGPT

Application: E-commerce Application

Generate Five User Stories for an E-commerce (end-to-end) application

- 1. User Registration** - As a new customer, I want to create an account so that I can securely access the e-commerce application.
- 2. Product Search** - As a customer, I want to search for products by name or keyword so that I can quickly find the products I need.
- 3. Shopping Cart** - As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.
- 4. Checkout and Payment** - As a customer, I want to complete checkout and make a payment so that I can place my order successfully.
- 5. Order Tracking** - As a customer, I want to track my order status so that I can know when my order will be delivered.

Convert One User Story into Acceptance Criteria

Selected User Story -> As a customer, I want to add products to my shopping cart so that I can purchase multiple products together.

Acceptance Criteria

1. Given a customer is viewing an available product, when they click Add to Cart, then the product should be added to the cart.

2. The cart should display the product name, unit price, quantity, and subtotal for each added product.
3. When the customer increases the quantity, the cart quantity and subtotal should update correctly.
4. When the customer decreases the quantity, the cart quantity and subtotal should update correctly.
5. When the customer clicks remove, the selected product should be removed from the cart.
6. The cart's total amount should be recalculated automatically whenever the quantity or products change.
7. If a product is unavailable, the customer should not be able to add it to the cart and should receive an appropriate message.

AI – Assisted Sprint Backlog

I asked ChatGPT to create a Sprint Backlog based on the above user stories. Here is the AI assisted Sprint Backlog-

ID	Task	Priority	Story Points
SB-01	Design user registration page	High	3
SB-02	Implement user registration	High	5
SB-03	Implement product search	High	5
SB-04	Create shopping cart functionality	High	8
SB-05	Implement quantity update	Medium	3
SB-06	Implement remove-from-cart functionality	Medium	3
SB-07	Develop checkout page	High	5
SB-08	Integrate payment functionality	High	8
SB-09	Implement order tracking	High	5
SB-10	Perform functional testing	High	5

Manual Review and Improvement

After reviewing the AI-generated Sprint Backlog, I identified several areas that could be improved.

Corrections ->

1. Added acceptance criteria validation tasks

The original backlog focused mainly on development tasks. I added QA activities such as test-case preparation and acceptance-criteria validation.

2. Separated development and testing

Instead of having only one general "functional testing" task, testing activities should be associated with individual features.

3. Improved task descriptions

For example:

"Implement quantity update " was too broad. I changed it into "Update product quantity"

4. Added negative testing

AI did not sufficiently cover scenarios such as:

- Invalid registration information
- Searching for a non-existing product
- Adding an unavailable product
- Invalid payment information
- Empty shopping cart checkout

5. Added priority based on business value

Core purchasing functionality should have higher priority than features such as order tracking.

For example: "Implement order tracking" will be "medium" priority as customer firstly place order successfully then use tracking feature.

What Did AI Do Well?

ChatGPT performed several tasks effectively:

- Assist me to create user stories using the standard "**As a..., I want..., so that...**" format.
- Assist me how I can convert a user story into multiple acceptance criteria.
- Generated a basic Sprint Backlog with priorities and story points.
- Helped identify different development activities that could be included in a Sprint.

What Corrections Did I Make?

I manually improved the AI output by:

- Making acceptance criteria more specific and testable.
- Adding QA and testing activities.
- Adding positive and negative test scenarios.
- Improving task descriptions.
- Adjusting priorities based on business importance.

Why Is Human Review Still Necessary?

AI can generate requirements and agile artifacts quickly, but human review is still necessary because AI may produce generic, incomplete, or unrealistic requirements.

A human QA/BA/Product Owner understands the actual business rules, customer expectations, technical limitations, and project priorities. AI may also miss important edge cases or create acceptance criteria that are difficult to test.

Therefore, AI should be treated as an assistant rather than a replacement for human judgment. Human review ensures that the final requirements are accurate, complete, testable, and aligned with the actual business needs.

Screenshots of AI Interaction

Assist me to create user story for **E-commerce application** using the standard "As a..., I want..., so that..." format with only one example.

As a customer, I want to search for products by name or keyword, so that I can quickly find the products I want to purchase.



Assist me how I can convert a user story into multiple acceptance criteria

To convert a **user story** into **multiple acceptance criteria**, first identify the specific conditions that must be true for the user story to be considered successfully completed.

Example User Story

As a customer, I want to add products to my shopping cart, so that I can purchase multiple products together.

Acceptance Criteria

1. The customer should be able to click the "Add to Cart" button for an available product.
2. The selected product should appear in the **shopping cart**.
3. The cart should display the **product name, price, and quantity**.
4. The customer should be able to **increase or decrease** the product quantity.
5. The customer should be able to **remove a product** from the cart.

- 5. The customer should be able to **remove a product** from the cart.
- 6. The cart **total price should update correctly** when the quantity changes.
- 7. An **unavailable product should not be added** to the cart.

Simple Method

Think of acceptance criteria as:

“What conditions must be satisfied for this user story to be accepted?”

For example:

User Story → Add product to cart

Break it into:

Add → Display → Quantity → Remove → Calculate Total → Handle Errors

This makes the user story **specific, testable, and easier for QA engineers to verify.**



Generated a Sprint Backlog

Generated Sprint Backlog

The AI generated a **Sprint Backlog** for the E-commerce application based on the previously created user stories. The backlog includes the tasks that should be completed during the Sprint, along with their **priority** and estimated **story points**.

Task	Priority	Story Points
Implement user registration	High	5
Implement product search	High	5
Add product to cart	High	3
Implement checkout process	High	5
Implement payment functionality	High	8

+

Ask anything

🧠 Think

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